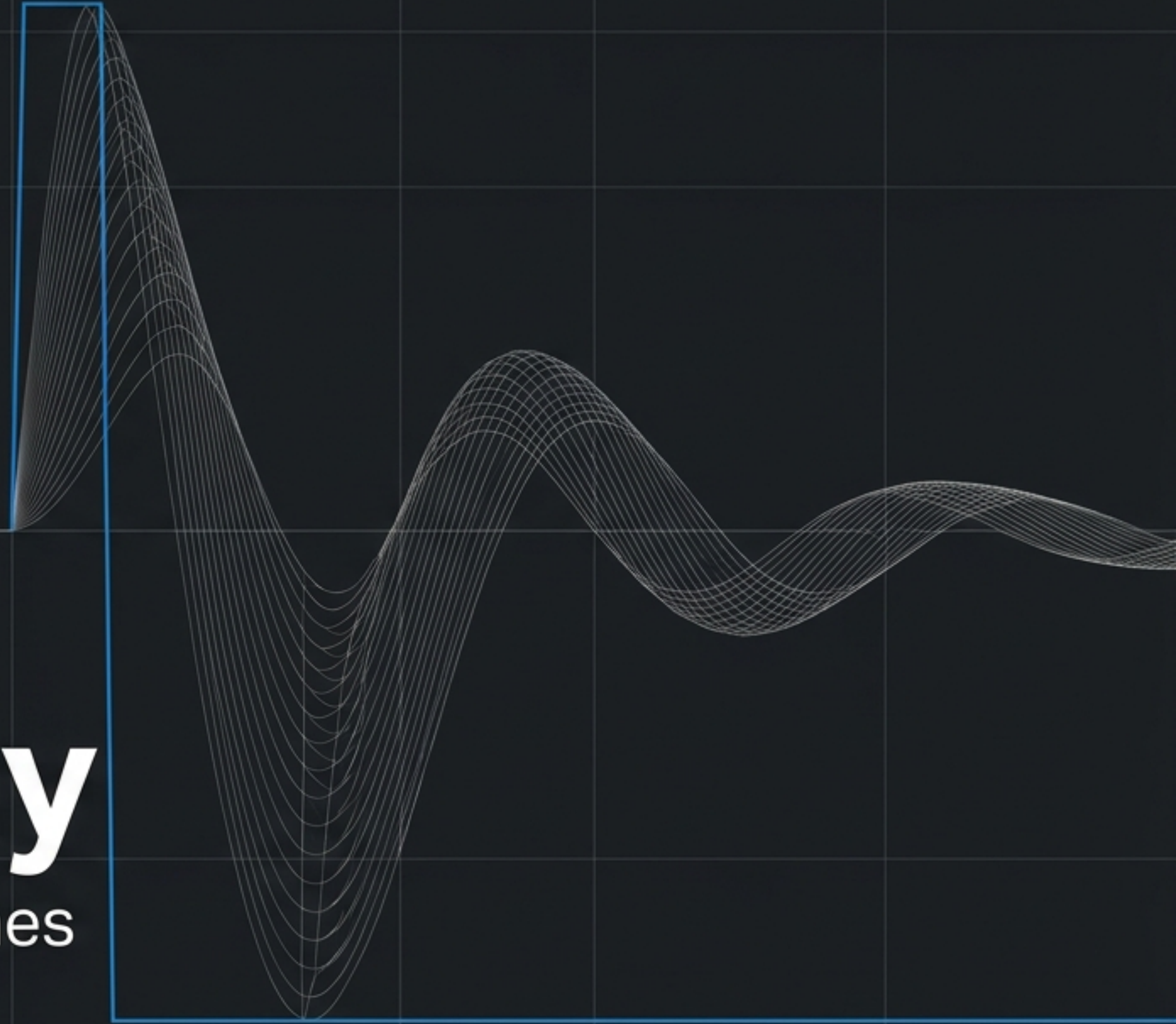


Beyond the Mercury Relay

Next-Generation Solid-State Switches
for Film Capacitor ESL Extraction

Strategic Analysis & Industrial Implementation Matrix



The switch's inductance is not a specification—it is your measurement floor.

$$L_{\text{total}} = \text{ESL} + L_{\text{loop}} + L_{\text{switch}}$$



The Bottleneck

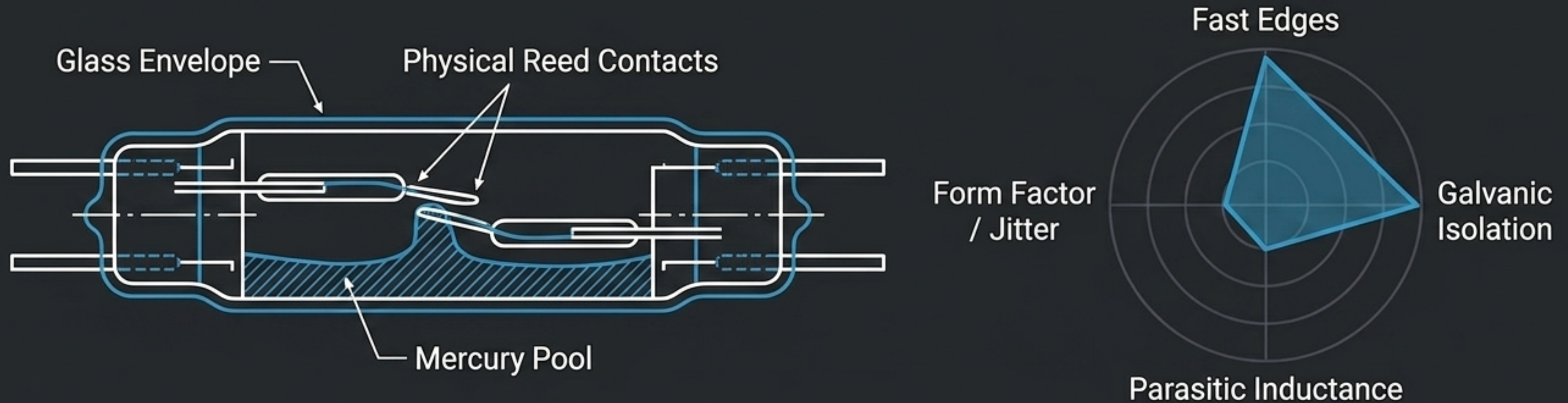
In nanohenry-range ESL extraction, switching transparency dictates accuracy. If the switch inductance exceeds the DUT inductance, the capacitor's characteristics are permanently masked by premature damping of the LC oscillation.

The Frequency Reality

A typical 1 μF film capacitor rings at roughly 1.6 MHz, with the critical first quarter-cycle around 150 ns. Sub-nanosecond edges are often overkill, but absolute minimal added inductance is strictly mandatory.

The limits of the laboratory dark horse: Mercury-Wetted Relays.

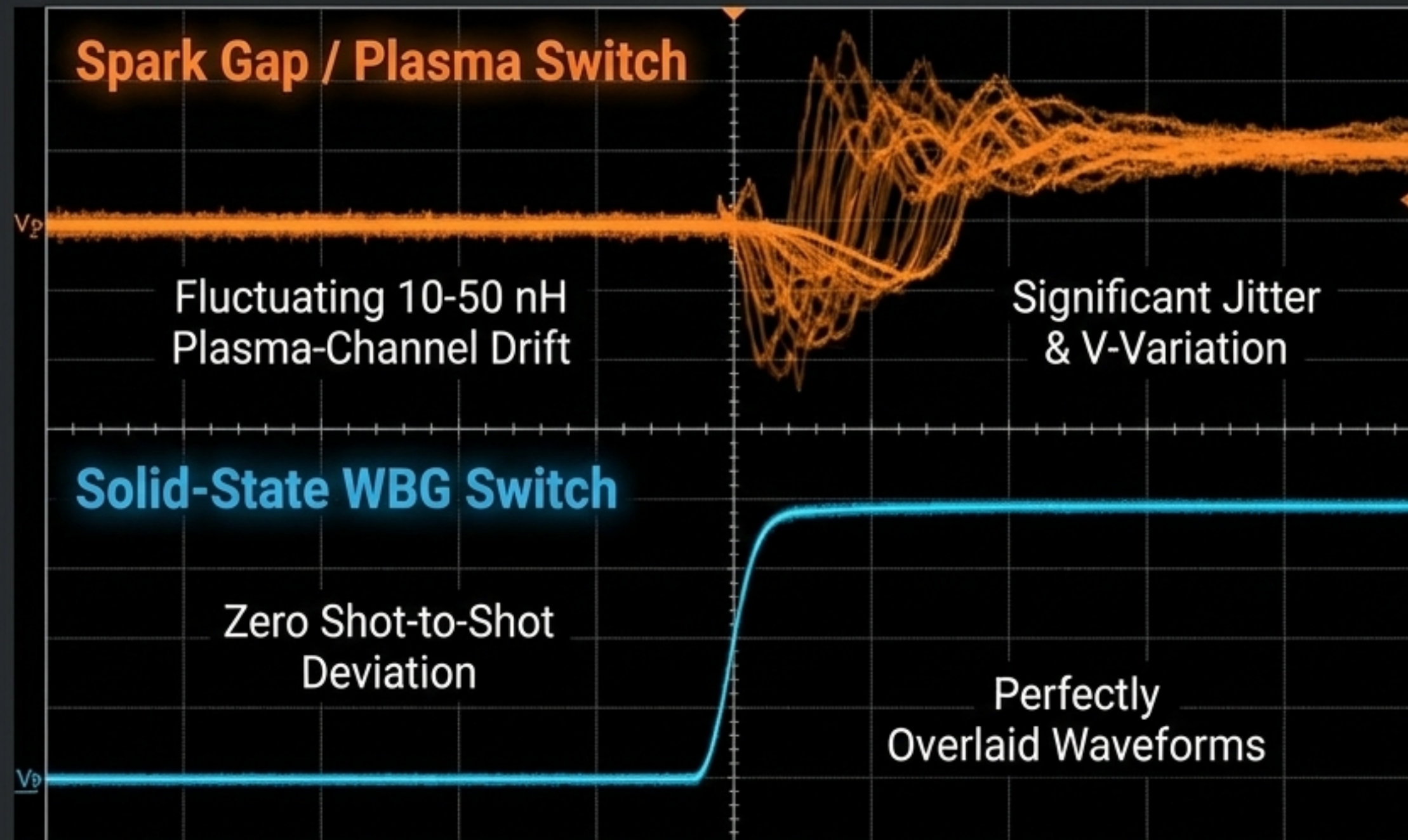
MWRs have long been prized for bounce-free, picosecond contact closure and inherent galvanic isolation. However, they are fundamentally incompatible with modern surface-mount/ultra-low ESL extraction.



- **The Lead Penalty:** Physical capsule layout unavoidably introduces 10–20 nH of parasitic lead inductance.
- **Mechanical Jitter:** Moving physical contacts introduce timing variability across repeated shots.
- **Orientation Sensitivity:** The internal mercury pool strictly dictates physical mounting angles, forcing bloated, sub-optimal test fixture layouts.

In nanohenry extraction, mathematical repeatability defeats raw speed.

The high-voltage literature often prioritizes sub-nanosecond transition speed. For ESL extraction, calibration repeatability is vastly more important.



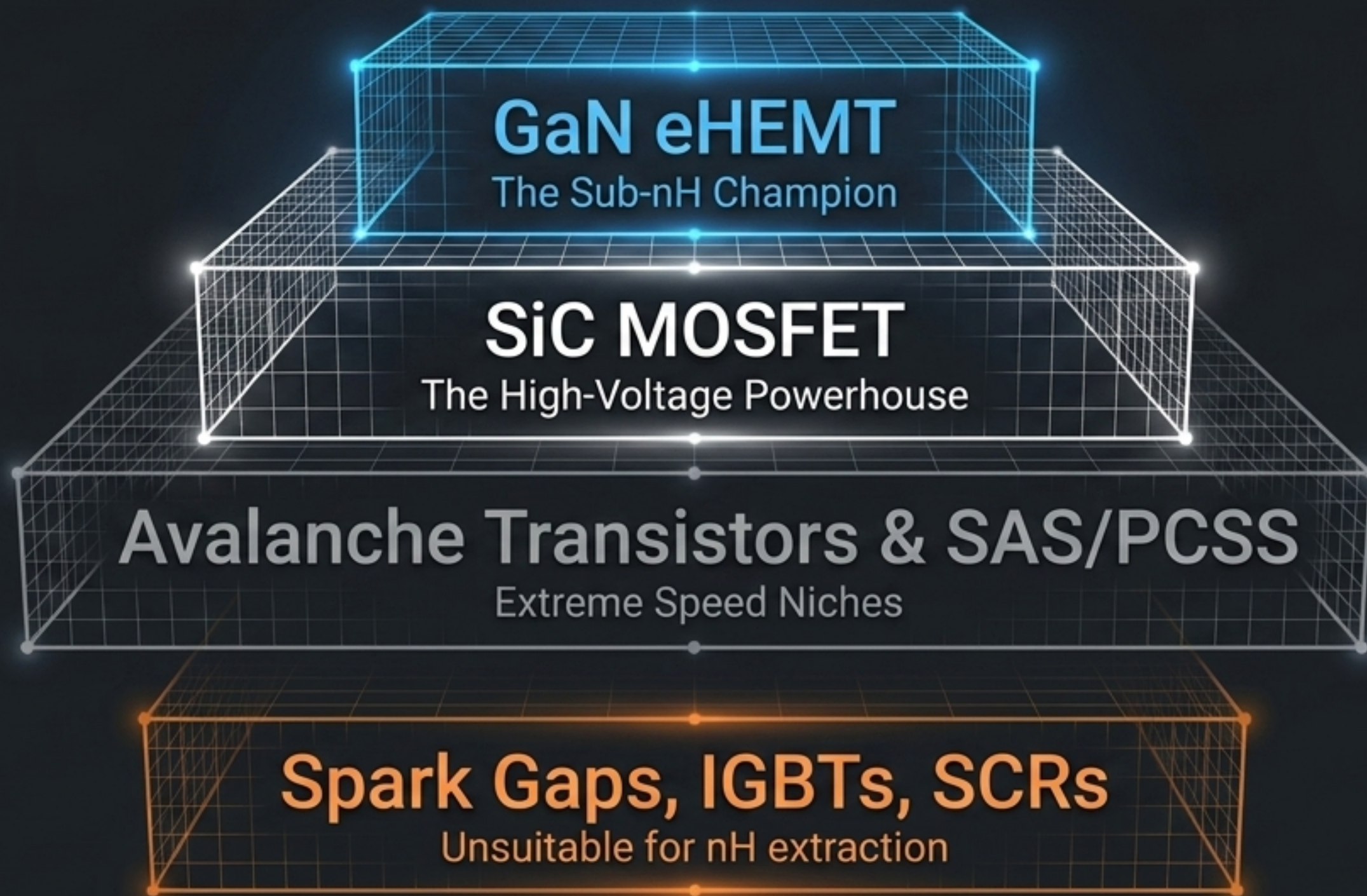
The De-embedding Imperative

Because L_{switch} must be mathematically **de-embedded** using a short-circuit reference shot, the switch's parasitic footprint must remain exactly the same shot-to-shot.

Solid-state devices achieve this tight grouping; mechanical and plasma variables **inherently drift**.

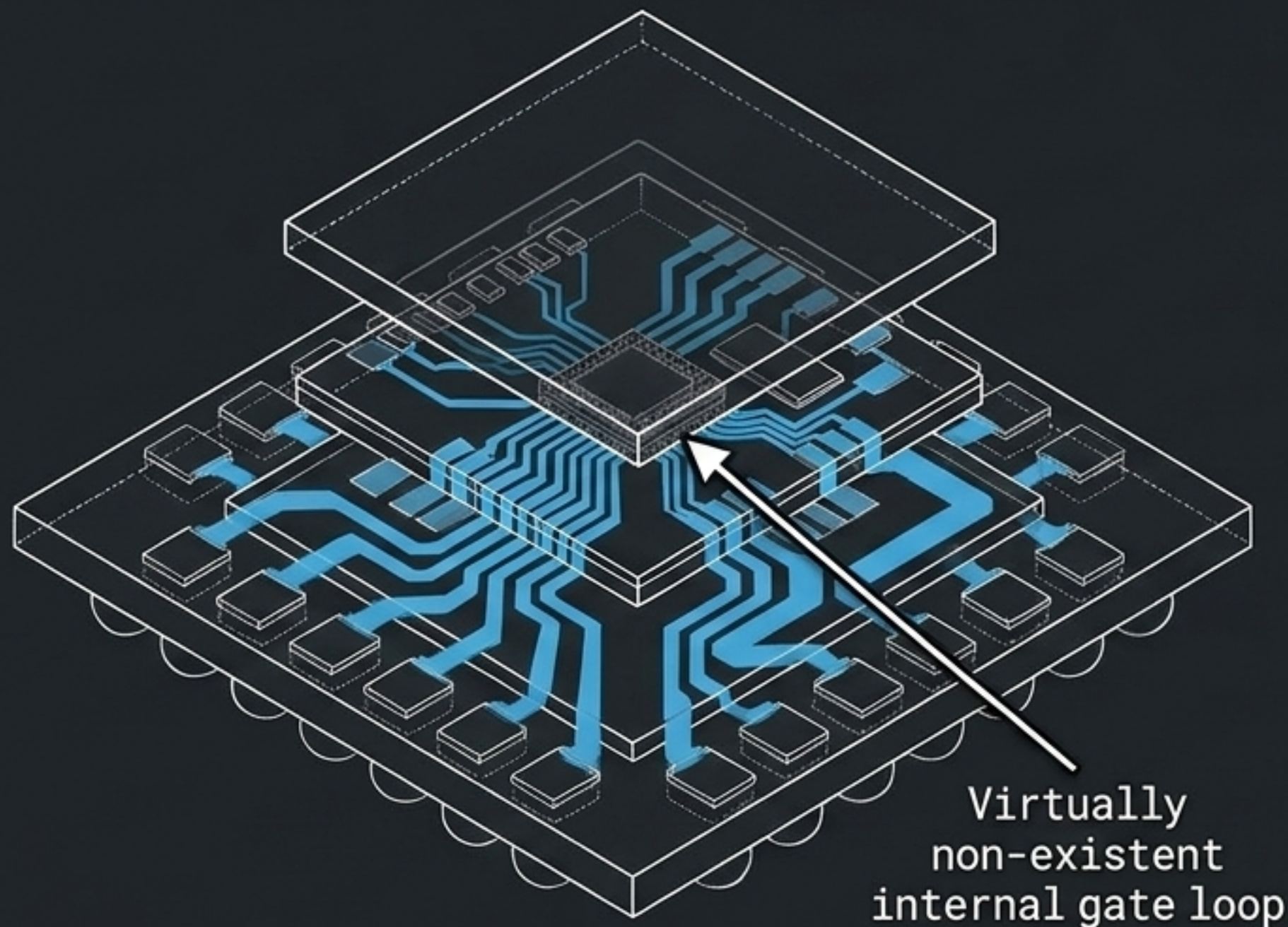
The solid-state hierarchy for ESL extraction.

Ranked strictly by parasitic footprint, transition speed, and industrial calibration repeatability.



#1 Ranked: GaN eHEMT (Gallium Nitride)

The ultimate all-around choice for moderate-voltage film capacitors.

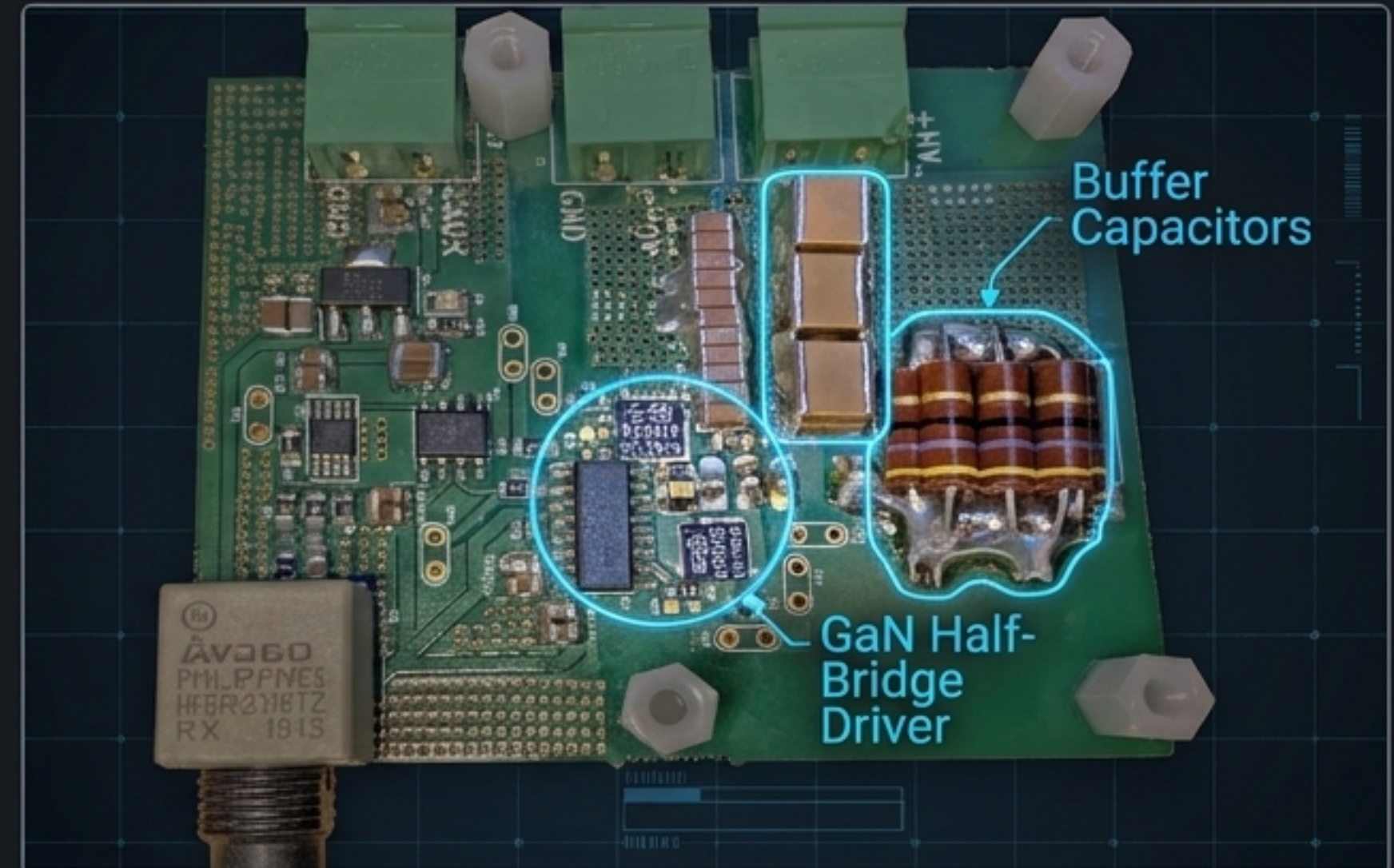
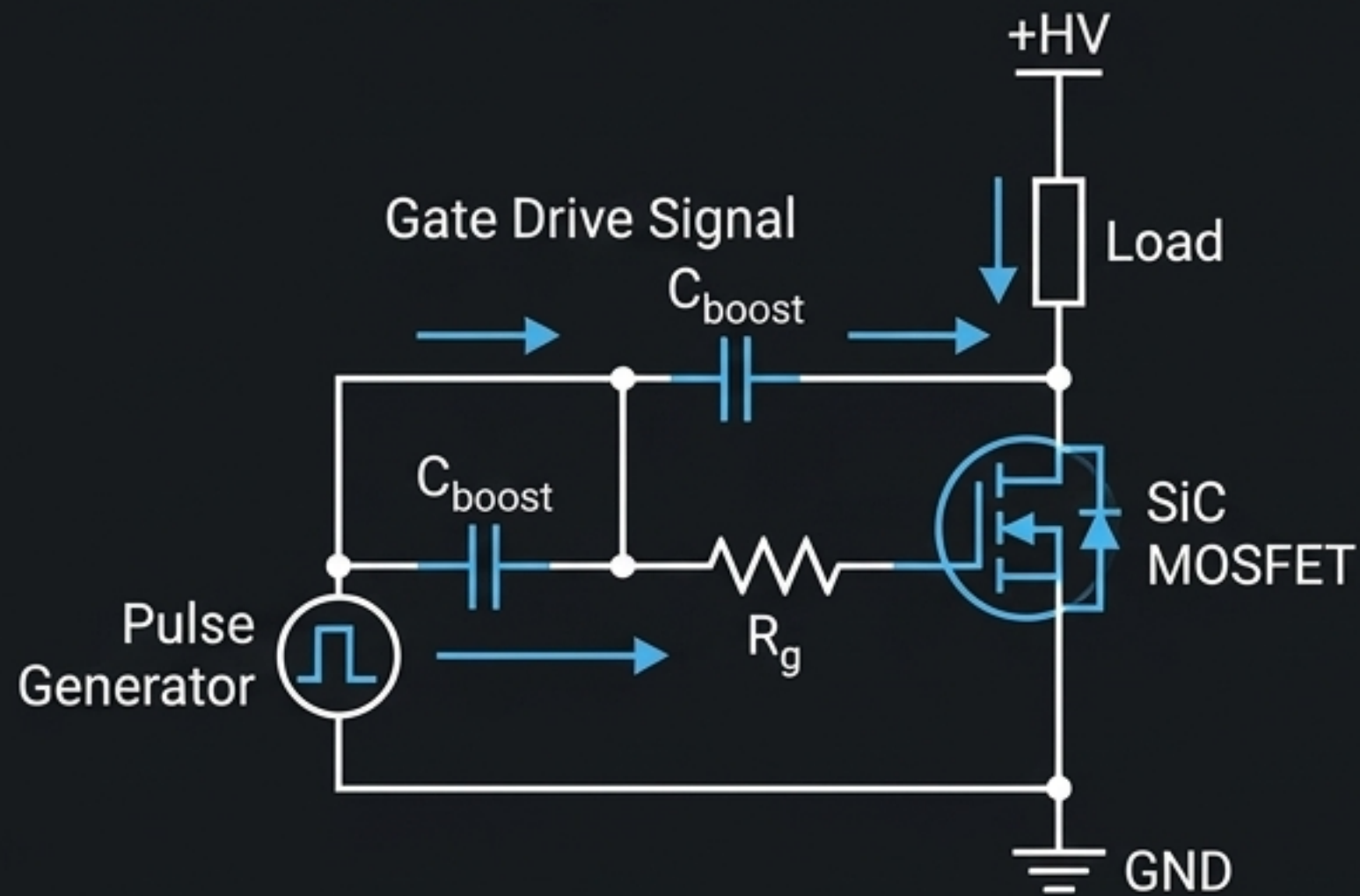


Transition Speed:	Sub-2 ns
Package Inductance:	<0.2–1.0 nH
Reverse Recovery (Q_{rr}):	Zero (vs. ~860 nC for Si)

Why it Wins: By collapsing the gate loop into a chip-scale package, integrated GaN FETs eliminate the gate-bounce and driver-ringing artifacts that plague discrete devices. Furthermore, zero Q_{rr} actively prevents secondary resonances from distorting the critical LC ring. Its microscopic residual inductance calibrates out flawlessly.

#2 Ranked: SiC MOSFET (The High-Voltage Powerhouse)

The necessary choice when the DUT voltage permanently exceeds GaN's comfortable limits, bridging high-voltage holdoff with ultra-fast extraction.



Data Metrics

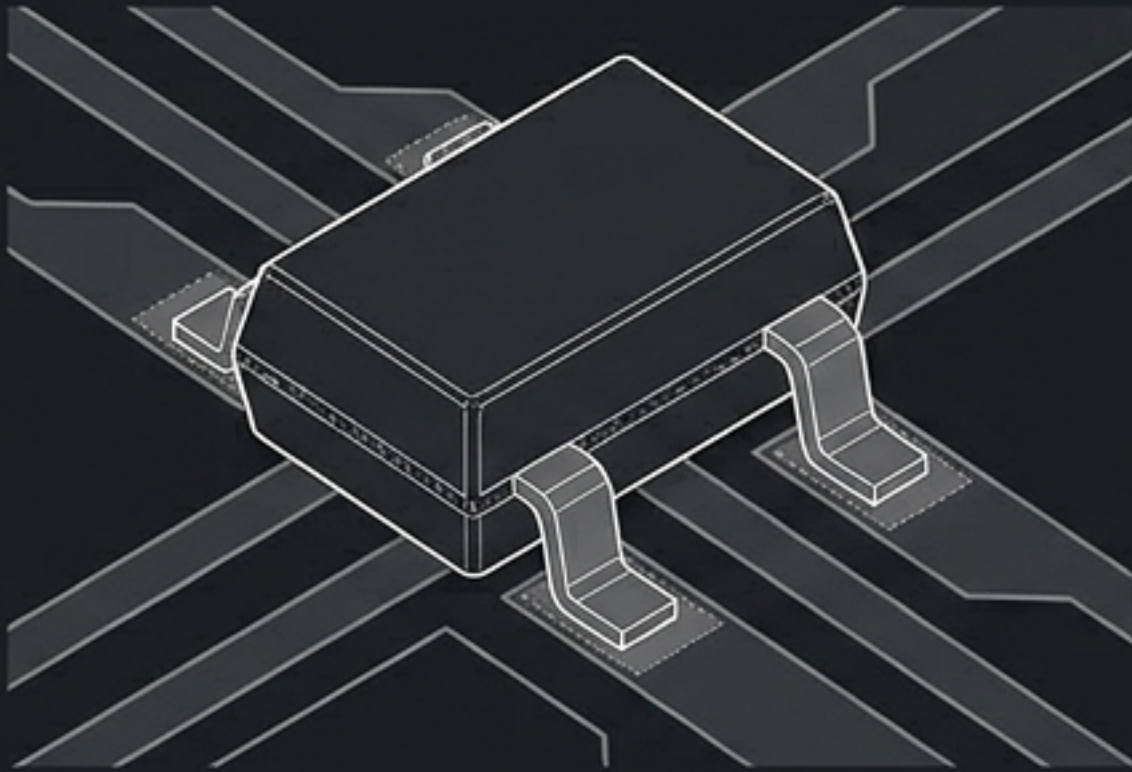
- Voltage Holdoff: >1.2 kV
- Transition Speed: <3 ns (with gate-boosting)
- Current Rise Rate: 38.7 kA/ μ s (at 356A load)

The Engineering Trick

Standard SiC turn-on is too slow. By utilizing an optimized gate-boosting circuit—applying a transient overvoltage to the gate via coupling capacitors—**SiC turn-on times** are violently slashed to <3 ns without damaging the die, generating massive di/dt capabilities.

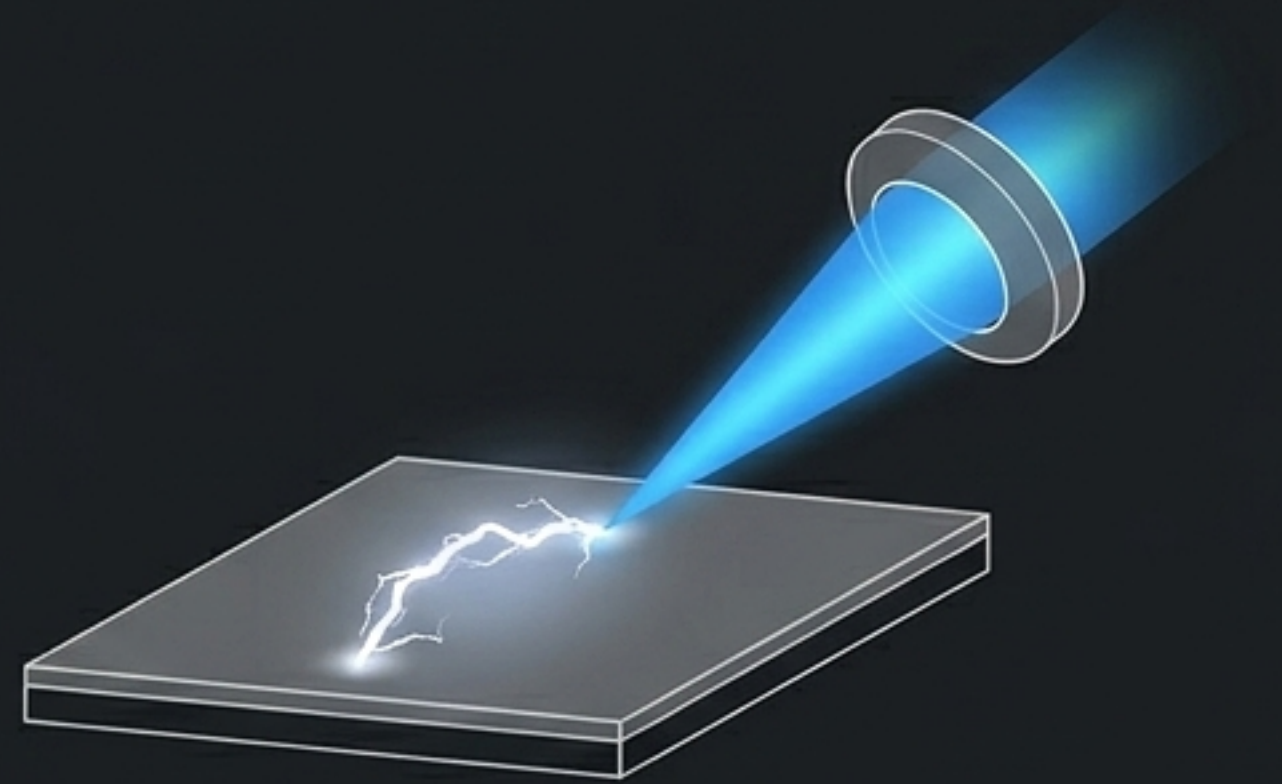
Honorable Mentions: When speed is the only metric.

Avalanche Transistors (e.g., 2N2369A)



- Unmatched 300–600 ps edges at low cost; ~2 nH package.
- **The Catch:** Extremely high device-to-device variability (~82% reach optimal speeds) and free-running jitter. Strictly relegated to low-energy, tiny-capacitor TDR-style work.

SAS & PCSS (Photoconductive Switches)

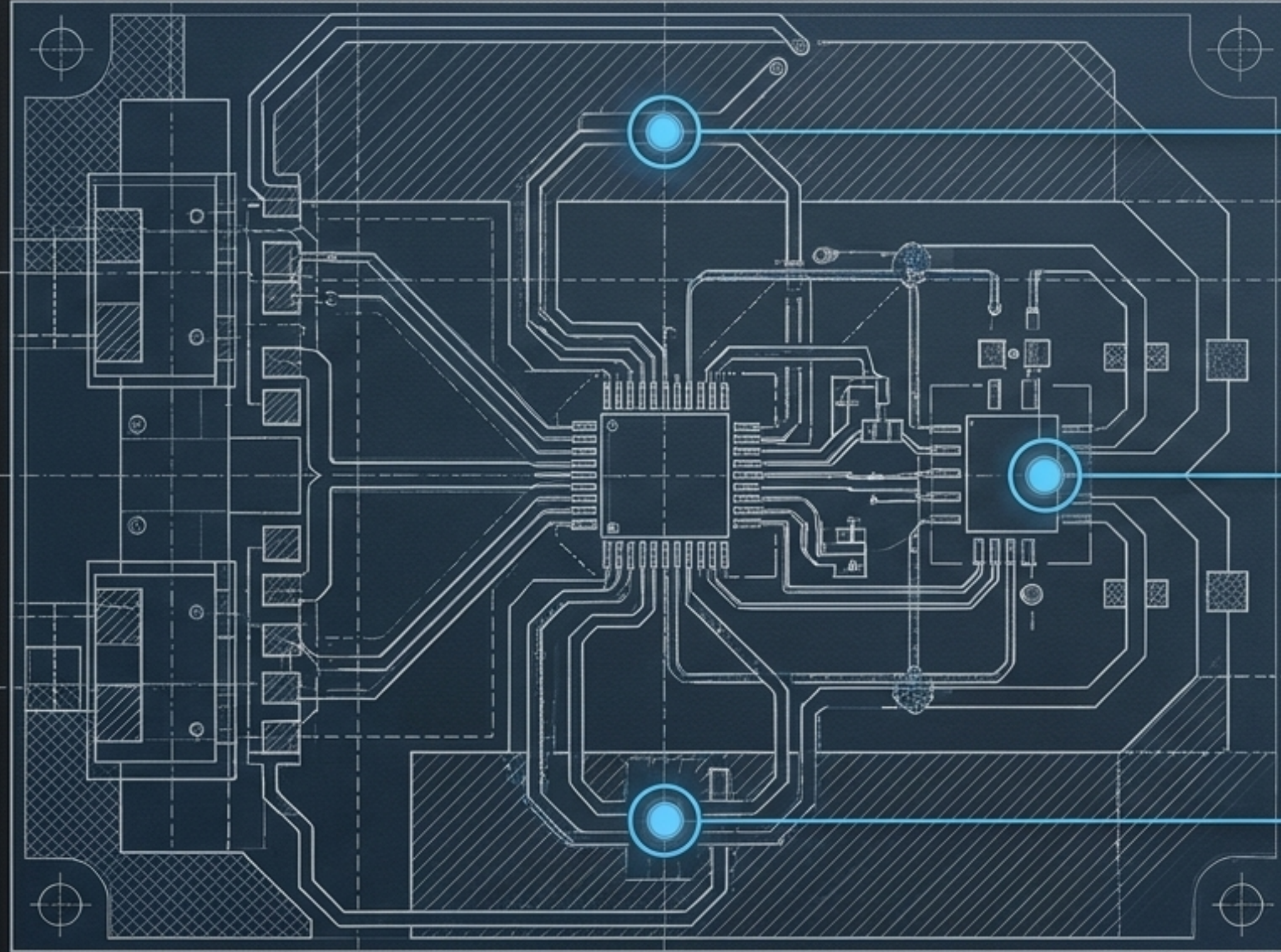


- **Performance:** 50-200 ps switching via laser-induced plasma injection or optical triggering. Zero mechanical jitter.
- **The Catch:** Extreme fixture complexity requiring dedicated optics and large laser trigger energies. Complete overkill for the 1.6 MHz ringing frequencies of standard 1 μ F film caps.

The Comprehensive ESL Switch Diagnostic Matrix.

Technology	Transition Speed	Package Inductance	Max Voltage	Repeatability (Gauge R&R)	Recommended Use
GaN eHEMT	<2 ns	<1 nH	<600V	Excellent	#1 Broad Industrial Use
SiC MOSFET	2–10 ns (Sub-3ns boosted)	6–8 nH	>1.2kV	Excellent	High-Voltage DUTs
MWR	1–5 ns	10–20 nH	>1kV	Poor–Fair	Legacy bench setups (Phase out)
Spark Gap	1–10 ns	10–50 nH	>10kV	Poor	Mega-energy banks only
Avalanche	300–600 ps	~2 nH	<100V	Variable	Sub-nF / TDR applications

The switch is only half the battle: Fixture Architecture.



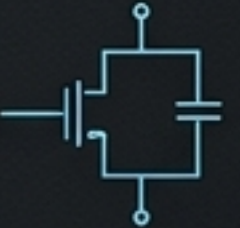
1. Cancel Magnetic Fields

Utilize vertical/multilayer stack-ups or parallel-plate geometries. This approach cuts loop inductance by ~70% compared to standard lateral layouts.



2. Kelvin-Connect the Switch

Strictly separate the gate-drive return from the power path to bypass package source inductance and prevent spurious gate ringing.



3. Non-Invasive Sensing

Discard in-line shunts that add parasitic series inductance. Utilize Rogowski coils or Pearson monitors to preserve the pristine low-inductance loop.



Strategic Imperatives for Industrial Measurement.

1

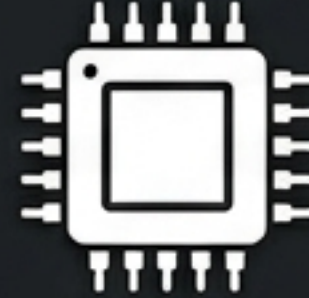


Retire the MWRs

Phase out mercury-wetted relays for surface-mount film capacitor testing.

The 10-20 nH lead penalty mathematically compromises modern sub-nanohenry ESL targets.

2



Standardize on WBG Solid-State

Adopt integrated GaN FETs for <600V testing and Gate-Boosted SiC for >1.2kV testing.

Optimize hardware for repeatable calibration, rather than chasing raw nanosecond speeds.

3



Mandate Short-Circuit Calibration

Always perform a reference shot using a low-inductance copper shim.

Single-shot regression fits are only valid if the switch footprint is mathematically zeroed out.